HW1: SIR and SEIR Model Implementation for Pandemic Spread.

Hypothesis for (i):

- What I'll do: I'll implement the SIR model with my own solver (ODE) and simulate it over 160 ~~day~~ days with $N = 2000$ people. Then, I'll find the day when I is at peak.
- What I'd expect & why: From $\frac{dI}{dt} = \beta SI - \gamma I$, with $S = 1995$ at the start, $\frac{dI}{dt} \approx 0.15$, which means that infections grow by around 15% per day. ~~Going~~ Going from 5 initial cases to several hundreds (5×1.15^{d}) would take about a month (30 days). The growth stops when $\beta SI = \gamma I \rightarrow S = \gamma/\beta = 800$. Since each person recovers at $\gamma = 0.1$ per day, I guess roughly less than half of those 1200 (2000-800) are still sick at peak.

Hypothesis for (ii):

- What I'll do: I'll implement the SEIR model with added parameters, simulate over 1500 days, and check whether one or several epidemic waves will come, describe the patterns (if any), and discuss about the effects
- What I'd expect and why: I expect several waves in this setting. Although birth and death rates are the same, all newborns would be included in S since they are not immune while deaths would be excluded from R. This means that the population stays the same but with more susceptible people, which will bring about more than one wave in 1500 days. Moreover, the exposed stage will delay the waves since it adds 4 days of latent period.

Hypothesis for (iii):

- What I'll do: I'll run the SEIR model on the given grids of β and γ, record the number of people at peak, and compare their effects.
- What I'd expect and why: I expect γ to affect the peak more than β. Both change R_0 equally, but γ also affects how long each person stays infectious. A larger γ means people recover faster, so it would make less people sick at the same time.